



10.2.1 A simple QR algorithm ¶

10.2.1 Simple QR algorithm, Part 1

fit width



We now morph the subspace iteration discussed in the last unit into a simple incarnation of an algorithm known as the **QR algorithm**. We will relate this algorithm to performing subspace iteration with an $m \times m$ (square) matrix so that the method finds all eigenvectors simultaneously (under mild conditions). Rather than starting with a random matrix V , we now start with the identity matrix. This yields the algorithm on the left in [Figure 10.2.1.1](#). We contrast it with the algorithm on the right.

$\hat{V} := I$	$V = I$
for $k := 0, \dots$	for $k := 0, \dots$
$(\hat{V}, \hat{R}) := \text{QR}(A\hat{V})$	$(Q, R) := \text{QR}(A)$
$\hat{A} = \hat{V}^H A \hat{V}$	$A = RQ$
endfor	$V = VQ$
	endfor

Figure 10.2.1.1. Left: Subspace iteration started with $\hat{V} = I$. Right: Simple QR algorithm.

The magic lies in the fact that the matrices computed by the QR algorithm are identical to those computed by the subspace iteration: Upon completion $\hat{V} = V$ and the matrix $\hat{A}$ on the left equals the (updated) matrix A on the right. To be able to prove this, we annotate the algorithm so we can reason about the contents of the matrices for each iteration.

$\hat{A}^{(0)} := A$	$A^{(0)} := A$
$\hat{V}^{(0)} := I$	$V^{(0)} := I$
$\hat{R}^{(0)} := I$	$R^{(0)} := I$
for $k := 0, \dots$	for $k := 0, \dots$
$(\hat{V}^{(k+1)}, \hat{R}^{(k+1)}) := \text{QR}(A\hat{V}^{(k)})$	$(Q^{(k+1)}, R^{(k+1)}) := \text{QR}(A^{(k)})$
$\hat{A}^{(k+1)} := \hat{V}^{(k+1)H} A \hat{V}^{(k+1)}$	$A^{(k+1)} := R^{(k+1)} Q^{(k+1)}$
$V^{(k+1)} := V^{(k)} Q^{(k+1)}$	
endfor	endfor

Let's start by showing how the QR algorithm applies unitary equivalence transformations to the matrices $A^{(k)}$.

Homework 10.2.1.1. Show that for the algorithm on the right

$$A^{(k+1)} = Q^{(k+1)H} A^{(k)} Q^{(k+1)}.$$

▼ [Solution](#)

The algorithm computes the QR factorization of $A^{(k)}$

$$A^{(k)} = Q^{(k+1)} R^{(k+1)}$$

after which

$$A^{(k+1)} := R^{(k+1)} Q^{(k+1)}$$

Hence

$$A^{(k+1)} = R^{(k+1)} Q^{(k+1)} = Q^{(k+1)H} A^{(k)} Q^{(k+1)}.$$

This last homework shows that $A^{(k+1)}$ is derived from $A^{(k)}$ via a unitary similarity transformation and hence has the same eigenvalues as does $A^{(k)}$. This means it also is derived from A via a (sequence of) unitary similarity transformation and hence has the same eigenvalues as does A .

We now prove these algorithms mathematically equivalent.

Homework 10.2.1.2. In the above algorithms, for all k ,

- $\hat{A}^{(k)} = A^{(k)}$.
- $\hat{R}^{(k)} = R^{(k)}$.

- $\widehat{V}^{(k)} = V^{(k)}$.

▼ [Hint](#) ▼ [Solution](#)

We will employ a proof by induction.

- Base case: $k = 0$

This is trivially true:

- $\widehat{A}^{(0)} = A = A^{(0)}$.
- $\widehat{R}^{(0)} = I = R^{(0)}$.
- $\widehat{V}^{(0)} = I = V^{(0)}$.

- Inductive step: Assume that $\widehat{A}^{(k)} = A^{(k)}$, $\widehat{R}^{(k)} = R^{(k)}$, and $\widehat{V}^{(k)} = V^{(k)}$. Show that $\widehat{A}^{(k+1)} = A^{(k+1)}$, $\widehat{R}^{(k+1)} = R^{(k+1)}$, and $\widehat{V}^{(k+1)} = V^{(k+1)}$.

From the algorithm on the left, we know that

$$A\widehat{V}^{(k)} = \widehat{V}^{(k+1)}\widehat{R}^{(k+1)}.$$

and

$$\begin{aligned} A^{(k)} &= \langle I, H. \rangle \\ \widehat{A}^{(k)} &= \langle \text{algorithm on left} \rangle \\ \widehat{V}^{(k)H} A \widehat{V}^{(k)} &= \langle \text{algorithm on left} \rangle \\ \widehat{V}^{(k)H} \widehat{V}^{(k+1)} \widehat{R}^{(k+1)} &= \langle \text{l.H.} \rangle \\ V^{(k)H} \widehat{V}^{(k+1)} \widehat{R}^{(k+1)} & \end{aligned} \quad (10.2.1)$$

But from the algorithm on the right, we know that

$$A^{(k)} = Q^{(k+1)} R^{(k+1)}. \quad (10.2.2)$$

Both (10.2.1) and (10.2.2) are QR factorizations of $A^{(k)}$ and hence, by the uniqueness of the QR factorization,

$$\widehat{R}^{(k+1)} = R^{(k+1)}$$

and

$$Q^{(k+1)} = V^{(k)H} \widehat{V}^{(k+1)}$$

or, equivalently and from the algorithm on the right,

$$\underbrace{V^{(k)} Q^{(k+1)}}_{V^{(k+1)}} = \widehat{V}^{(k+1)}.$$

This shows that

- $\widehat{R}^{(k+1)} = R^{(k+1)}$ and
- $\widehat{V}^{(k+1)} = V^{(k+1)}$.

Also,

$$\begin{aligned} \widehat{A}^{(k+1)} &= \langle \text{algorithm on left} \rangle \\ \widehat{V}^{(k+1)H} A \widehat{V}^{(k+1)} &= \langle \widehat{V}^{(k+1)} = V^{(k+1)} \rangle \\ V^{(k+1)H} A V^{(k+1)} &= \langle \text{algorithm on right} \rangle \\ Q^{(k+1)H} V^{(k)H} A V^{(k)} Q^{(k+1)} &= \langle \text{l.H.} \rangle \\ Q^{(k+1)H} \widehat{V}^{(k)H} A \widehat{V}^{(k)} Q^{(k+1)} &= \langle \text{algorithm on left} \rangle \\ Q^{(k+1)H} \widehat{A}^{(k)} Q^{(k+1)} &= \langle \text{l.H.} \rangle \\ Q^{(k+1)H} A^{(k)} Q^{(k+1)} &= \langle \text{last homework} \rangle \\ A^{(k+1)} & \end{aligned}$$

- By the Principle of Mathematical Induction, the result holds.

The QR factorization is unique, provided the diagonal elements of R are taken to be positive.

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Homework 10.2.1.3. In the above algorithms, show that for all k

- $V^{(k)} = Q^{(0)} Q^{(1)} \dots Q^{(k)}$.
- $A^k = V^{(k)} R^{(k)} \dots R^{(1)} R^{(0)}$. (Note: A^k here denotes A raised to the k th power.)

Assume that $Q^{(0)} = I$.

▼ [Solution](#)

We will employ a proof by induction.

- Base case: $k = 0$

$$\underbrace{A^0}_I = \underbrace{V^{(0)}}_I \underbrace{A^0}_I = \underbrace{R^{(0)}}_I.$$

- Inductive step: Assume that $V^{(k)} = Q^{(0)} \dots Q^{(k)}$ and $A^k = V^{(k)} R^{(k)} \dots R^{(0)}$. Show that $V^{(k+1)} = Q^{(0)} \dots Q^{(k+1)}$ and $A^{k+1} = V^{(k+1)} R^{(k+1)} \dots R^{(0)}$.

$$V^{(k+1)} = V^{(k)} Q^{(k+1)} = Q^{(0)} \dots Q^{(k)} Q^{(k+1)}.$$

by the inductive hypothesis.

Also,

$$\begin{aligned} A^{k+1} &= \text{< definition >} \\ A A^k &= \text{< inductive hypothesis >} \\ A V^{(k)} R^{(k)} \dots R^{(0)} &= \text{< inductive hypothesis >} \\ A \widehat{V}^{(k)} R^{(k)} \dots R^{(0)} &= \text{< left algorithm >} \\ \widehat{V}^{(k+1)} \widehat{R}^{(k+1)} R^{(k)} \dots R^{(0)} &= \text{< } V^{(k+1)} = \widehat{V}^{(k+1)}; R^{(k+1)} = \widehat{R}^{(k+1)} \text{ >} \\ V^{(k+1)} R^{(k+1)} R^{(k)} \dots R^{(0)}. \end{aligned}$$

- By the Principle of Mathematical Induction, the result holds for all k .

This last exercise shows that

$$A^k = \underbrace{Q^{(0)} Q^{(1)} \dots Q^{(k)}}_{\text{unitary } V^{(k)}} \underbrace{R^{(k)} \dots R^{(1)} R^{(0)}}_{\text{upper triangular } \widetilde{R}^{(k)}}$$

which exposes a QR factorization of A^k . Partitioning $V^{(k)}$ by columns

$$V^{(k)} = \left(\begin{array}{c|c} v_0^{(k)} & \dots & v_{m-1}^{(k)} \end{array} \right)$$

we notice that applying k iterations of the Power Method to vector e_0 yields

$$A^k e_0 = V^{(k)} \widetilde{R}^{(k)} e_0 = V^{(k)} \tilde{\rho}_{0,0}^{(k)} e_0 = \tilde{\rho}_{0,0}^{(k)} V^{(k)} e_0 = \tilde{\rho}_{0,0}^{(k)} v_0^{(k)},$$

where $\tilde{\rho}_{0,0}^{(k)}$ is the $(0,0)$ entry in matrix $\widetilde{R}^{(k)}$. Thus, the first column of $V^{(k)}$ equals a vector that would result from k iterations of the Power Method. Similarly, the second column of $V^{(k)}$ equals a vector that would result from k iterations of the Power Method, but orthogonal to $v_0^{(k)}$.

10.2.1 Simple QR algorithm, Part 2

fit width



We make some final observations:

- $A^{(k+1)} = Q^{(k)H} A^{(k)} Q^{(k)}$. This means we can think of $A^{(k+1)}$ as the matrix $A^{(k)}$ but viewed in a new basis (namely the basis that consists of the column of $Q^{(k)}$).
- $A^{(k+1)} = (Q^{(0)} \dots Q^{(k)})^H A Q^{(0)} \dots Q^{(k)} = V^{(k)H} A V^{(k)}$. This means we can think of $A^{(k+1)}$ as the matrix A but viewed in a new basis (namely the basis that consists of the column of $V^{(k)}$).
- In each step, we compute

$$(Q^{(k+1)}, R^{(k+1)}) = QR(A^{(k)})$$

which we can think of as

$$(Q^{(k+1)}, R^{(k+1)}) = QR(A^{(k)} \times I).$$

This suggests that in each iteration we perform one step of subspace iteration, but with matrix $A^{(k)}$ and $V = I$:

$$(Q^{(k+1)}, R^{(k+1)}) = QR(A^{(k)} V).$$

- The insight is that the QR algorithm is identical to subspace iteration, except that at each step we reorient the problem (express it in a new basis) and we restart it with $V = I$.

Homework 10.2.1.4. Examine

`Assignments/Week10/matlab/SubspaceIterationAllVectors.m` , which implements the subspace iteration in [Figure 10.2.1.1](#) (left). Examine it by executing the script in `Assignments/Week10/matlab/test_simple_QR_algorithm.m` .

▼ [Solution](#)

Discuss what you observe online with others!

Homework 10.2.1.5. Copy

`Assignments/Week10/matlab/SubspaceIterationAllVectors.m` into `SimpleQRAlg.m` and modify it to implement the algorithm in [Figure 10.2.1.1](#) (right) as

```
function [ Ak, V ] = SimpleQRAlg( A, maxits, illustrate, delay )
```

Modify the appropriate line in `Assignments/Week10/matlab/test_simple_QR_algorithms.m` , changing (0) to (1), and use it to examine the convergence of the method.

What do you observe?

▼ [Solution](#)

- `Assignments/Week10/answers/SimpleQRAlg.m`

Discuss what you observe online with others!